Computer Networks - Midterm Cheat Sheet

Chapters 1-5 . Dordal, An Introduction to Computer Networks . every exam question is a textbook exercise . next_hop in the exercises = neighbor form (name the neighbor, not the port number).

1. Datagram Forwarding (Ch. 1)

- Forwarding table row = <destination, next_hop>. Directly attached destination -> direct / local.
- Flood = send out every port except the arrival port (used when the destination is unknown, or for broadcast).
- Build a table: for each destination take the shortest path; next_hop = first switch on that path. If two are tied, either is fine - just indicate it.
- Default row <default, next_hop> catches the rest. Delete any explicit row whose next_hop equals the default's.

> Routing loops: only two shapes - two adjacent switches bouncing a packet, or a real topological cycle. A linear loop of length > 1 is impossible (next hop is a function; a repeat traps the packet). Defenses: TTL (decrement per hop, drop at 0) and split horizon.

- Weighted links: pick the path of least total weight. Inverse problem (make this route optimal): weight 1 everywhere, one or two bad links weigh 10.
- Subnets: each network 200.0.x/24 is one route row - local if directly attached, else the neighbor router.

2. Ethernet Basics (Ch. 2)

- CSMA/CD: listen; if idle, transmit; on a collision send a jam signal and back off.

> Exponential backoff: after the n-th collision, choose k in [0, 2^n - 1] slot times and retransmit 1 + k slots later. Round n caps at 10; give up after 16 collisions.

- Slot time = 512 bit-times = 51.2 microseconds.
- Minimum frame = 64 bytes (so collisions are detected within the slot time).
- Maximum payload = 1500 bytes (max frame 1518 B).

Learning switch - every frame

- Learn the source address -> its arrival port.
- Destination known -> forward that one port; unknown -> flood; broadcast -> flood.
- Process frames in order; update every switch that saw each frame (a known destination stops the frame early).
- Loop without STP: layer 2 has no TTL -> unknown/broadcast frames flood forever.
- VLAN: logical color / ID per port; trunk frames tagged 802.1Q.

3. Advanced Ethernet (Ch. 3)

Spanning Tree (STP)

- Root = switch with the lowest ID.
- Each non-root switch keeps its least-cost path to the root; ties broken by the lowest-ID neighbor.
- Everything else is pruned / blocked -> loop-free tree. Switches exchange BPDUs.

SDN / OpenFlow

- Switch = flow tables of match -> action.
- Unknown destination -> report to the controller; the controller learns the source and installs source -> arrival port; the switch then floods.
- Destination known -> forward normally, no report.

4. Wireless LANs (Ch. 4)

- Hidden node: two senders each reach the receiver but cannot hear each other -> their transmissions collide at the receiver.
- CSMA/CA (Wi-Fi): no collision detection; random backoff, DIFS > SIFS, ACK/CTS after SIFS, RTS/CTS for hidden nodes.
- ACK without sensing is OK: it is small and follows a finished transmission; a collision only matters if a node reads neither signal.

> Contention loss: 802.11g slot = 9 microseconds; 1 kB at 54 Mbps ~ 151 microseconds. Fraction of bandwidth lost = T_cont / (T_cont + T_tx). (avg 64 slots -> 576 us -> ~79% lost.)

- WiMAX/LTE are scheduled (base station allocates); Wi-Fi listens and defers.
- WiMAX ranging: 33 m/s x 60 s = 1980 m -> 1980/300 ~ 6.6 us early/late; guard = 2x -> 5000 senders/s ~ 66 ms/s ~ 6.6%.
- Two-ray ground: reflected path r' ~ r(1 + 2h^2/r^2); subtract the 180-degree-shifted signals -> amplitude ~ 1/r^2 -> power ~ 1/r^4.
- WPA2 4-way handshake: a replay fails when the AP sends a fresh nonce - the old nonce gives the wrong MIC.

5. Other LANs (Ch. 5)

VPN / tunneling

- Private-addressed packets are encapsulated inside normal Internet packets to a VPN gateway (e.g. eth0 public, ppp0 tunnel).
- All else via ppp0 -> traffic to a local/public address is forced into the tunnel.
- All traffic via ppp0 -> the tunnel's own packets get tunneled -> recursion / loop. A split tunnel must exclude the gateway address.

Token Ring / Bus

- Only the token holder transmits; average passes over N stations = N/2; order can be set by sorting MAC addresses.
- 64 B token at 10 Mbps = 51.2 us; x 20 = 1.024 ms.

ATM / AAL-5

- Cell = 53 B (5 B header + 48 B payload). SM90: N cells + 1 XOR cell -> recover any single loss.
- With XOR = 21 cells (no retransmit); without ~ 20 + 400p. Equal at 21 = 20 + 400p -> p = 0.25%.

Virtual Circuits

- VCI table columns: VCI_in | port_in | VCI_out | port_out. A port = the node at the other end; VCIs are per-link (reusable).
- Trace a VC: start at the ingress row -> follow VCI_out / port_out to the next switch -> match its VCI_in / port_in -> repeat to the endpoint. A switch visited twice has two rows.

Quick numbers and formulas

Quantity	Value
Ethernet slot time (10 Mbps)	512 bit-times = 51.2 us
Min frame / max payload	64 B / 1500 B
802.11g slot time	9 us
Transmit time	bits / data-rate
Contention loss	T_cont / (T_cont + T_tx)
WiMAX delay error	(speed x interval) / c
Two-ray received power	proportional to 1/r^4
ATM cell	5 B header + 48 B
Token wait (N stations)	(N/2) x token-tx-time

Cross-check every worked example against the book's Ch. 34 'Selected Solutions' and the exercise-numbering conversion table.